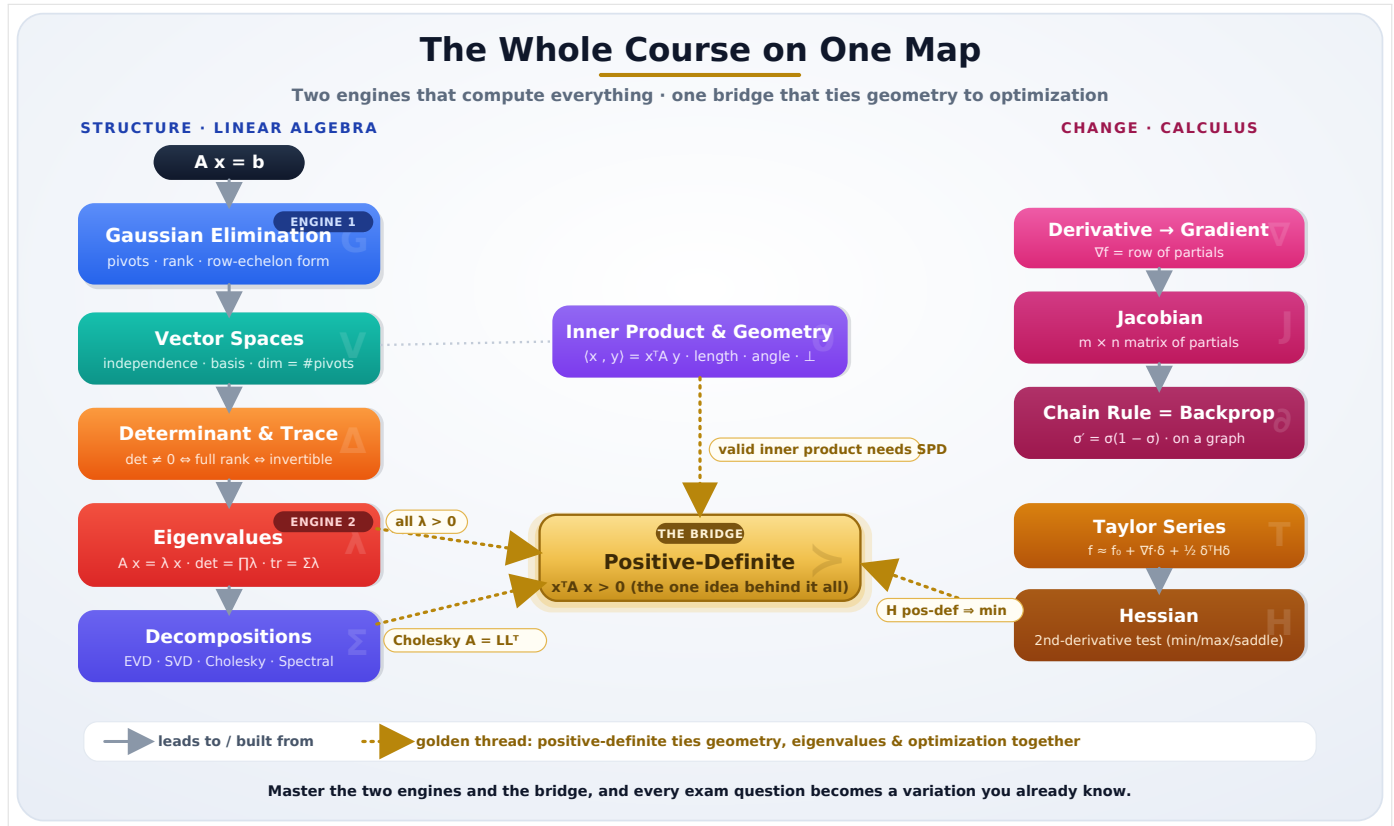


MFML — Concept Map & Formula Guide

Mathematical Foundations for Machine Learning (AIMLCZC416). The course is really just **two engines and one bridge** — learn how the pieces fit, and the formulas hang neatly off that picture.

START HERE

The one picture that connects everything



Read it like a story: solve $Ax = b$ and elimination hands you the **pivots and rank**; those open up vector spaces, determinants and eigenvalues, which build every decomposition. The gold lines point to the one idea — **positive-definite** — that keeps showing up in geometry, eigenvalues and optimization.

WHY IT STICKS

Four golden threads that run through the whole course

1 · Pivots & rank are the answer key

Gaussian elimination tides any matrix into a staircase. The leading entries are the **pivots**; their count is the **rank**. Knowing the rank answers most of Lectures 1–4: how many solutions a system has, whether vectors are independent, the dimension of a space, and whether a matrix can be inverted.

Links: runs through linear systems, spaces & eigenvalues

2 · Positive-definite wears three hats

One simple check, $x^T A x > 0$ (**positive-definite**), keeps returning in three forms: it makes an **inner product valid**, it lets you factor $A = LL^T$ (**Cholesky**), and it tells you a critical point is a **minimum**. Test it once, recognise it everywhere.

Links: inner products, Cholesky & optimization

3 · Trace & det are eigenvalues in disguise

For any square matrix, the **trace** (sum of the diagonal) $= \sum \lambda$ and the **determinant** $= \prod \lambda$. So eigenvalues secretly decide invertibility, what powers look like, and how a matrix splits apart. Symmetric $\Rightarrow$ real λ and perpendicular eigenvectors.

Links: determinants, eigenvalues & decompositions

4 · The chain rule scales to deep nets

A **gradient** is a row of slopes (partial derivatives). A **Jacobian** stacks gradients ($m \times n$). Link them with the chain rule and you get **backpropagation** — the chain rule run backwards through a network. Same idea, from $f'(x)$ to deep learning.

Links: gradients, Jacobians & backprop

THE FORMULAS THAT MATTER

Everything you actually need, one card per topic

Linear Systems L1

- **3 row ops:** swap $R_i \leftrightarrow R_j$; scale $R_i \rightarrow cR_i$ ($c \neq 0$); add $R_i \rightarrow R_i + cR_j$.
- **Solution type** ($r = \text{rank } A, \tilde{r} = \text{rank}[A|b], n$): $r < \tilde{r}$ none; $r = \tilde{r} = n$ unique; $r = \tilde{r} < n \infty$ ($n - r$ free).
- **Full answer:** $\mathbf{x} = \mathbf{x}_p + \sum \lambda_i \mathbf{x}_{\text{null}}$.
- **Square A :** invertible $\iff \det A \neq 0 \iff$ full rank; inverse via $[A|I] \rightarrow [I|A^{-1}]$.
- **Handy:** $(AB)^T = B^T A^T$, $(AB)^{-1} = B^{-1} A^{-1}$, 2×2 inv $= \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

Links: same pivots decide independence, basis & invertibility.

Remember: ranks differ $\rightarrow$ none; $= n \rightarrow$ one; $< n \rightarrow$ many.

Vector Spaces L2

- **Subspace test:** contains $\mathbf{0}$ and closed: $\lambda x + \mu y$ stays inside.
- **Independent** $\iff$ only $\sum \lambda_i x_i = \mathbf{0}$ is all-zero $\iff$ every column is a pivot.
- **Basis** = independent + spanning; dim = number of basis vectors = rank.
- **Size rule:** $U \subseteq V \Rightarrow \dim U \leq \dim V$ (equal $\iff U = V$).
- **Quick:** a set with $\mathbf{0}$ is dependent; any $m > n$ vectors in $\mathbb{R}^n$ are dependent.

Links: it is Gaussian elimination again — pivot columns are the basis.

Remember: columns $\rightarrow$ row-echelon $\rightarrow$ pivots are the keepers.

Inner Products & Geometry L3

- **Norm rules:** $\|\lambda x\| = |\lambda| \|x\|$, $\|x + y\| \leq \|x\| + \|y\|$, $\|x\| \geq 0$; $\|x\|_2 = \sqrt{\sum x_i^2}$.
- **Inner product:** $\langle x, y \rangle = x^T A y$ (A SPD); $\|x\| = \sqrt{\langle x, x \rangle}$; $d = \|x - y\|$.
- **Angle:** $\cos \theta = \frac{\langle x, y \rangle}{\|x\| \|y\|}$; $\perp \iff \langle x, y \rangle = 0$; C-S $|\langle x, y \rangle| \leq \|x\| \|y\|$.
- **Orthogonal** $Q^T Q = I$; Gram-Schmidt $u_k = v_k - \sum_{j < k} \frac{\langle v_k, u_j \rangle}{\langle u_j, u_j \rangle} u_j$.
- **Complex:** conjugate transpose; Hermitian $A^H = A$, $\langle u, v \rangle = u^H v$.

Links: a valid inner product needs the positive-definite bridge.

Remember: $\cos \theta = \langle x, y \rangle / (\|x\| \|y\|)$ — “dot over norms”.

Determinant, Trace & Eigenvalues L4

- **Eigen:** $Ax = \lambda x \iff \det(A - \lambda I) = 0$; eigenspace $= \text{null}(A - \lambda I)$.
- **Det:** $2 \times 2 = ad - bc$; **triangular/diagonal** $= \prod$ **diagonal**; row swap $\times(-1)$, add multiple unchanged; $\det(A^n) = (\det A)^n$.
- **Golden:** trace $= \sum \lambda$, $\det = \prod \lambda$; $\text{tr}(AB) = \text{tr}(BA)$.
- **Symmetric:** real λ , orthogonal eigvecs; spectral $A = Q \Lambda Q^T$.
- **Cholesky:** $A = LL^T$, $l_{11} = \sqrt{a_{11}}$, $l_{21} = \frac{a_{21}}{l_{11}}$, $l_{22} = \sqrt{a_{22} - l_{21}^2}$.
- **Multiplicity:** algebraic $\geq$ geometric ($\dim E_\lambda$).

Links: the structure engine — powers & every decomposition depend on λ .

Remember: Trace $= \sum \lambda$, Det $= \prod \lambda$; triangular $\Rightarrow \lambda$ on the diagonal.

Decompositions: SVD & friends L5

- **Diagonalize:** $D = P^{-1}AP$; EVD $A = PDP^{-1}$.
- **SVD (any A):** $A = U \Sigma V^T$; V = eigvecs of $A^T A$, U = eigvecs of AA^T .
- **Singular values:** $\sigma_i = \sqrt{\lambda_i(A^T A)}$ (descending); $u_i = \frac{1}{\sigma_i} A v_i$.
- **Rank- k :** $A = \sum_i \sigma_i u_i v_i^T$ — first k terms = best approximation.
- **Spectral norm:** $\|A\|_2 = \sigma_1$ (largest singular value).

Links: all built from eigenvalues; Cholesky leans on positive-definite.

Remember: right V from $A^T A$, left U from AA^T , $\sigma = \sqrt{\text{eig}}$.

Gradients, Jacobians & Backprop L6-7

- **Shapes:** gradient $\nabla f = [\partial f / \partial x_i]$ (row $1 \times n$); Jacobian $[\partial f_i / \partial x_j]$ ($m \times n$).
- **Rules:** $(fg)' = f'g + fg'$, quotient, chain $(g \circ f)' = g'(f)f'$.
- **Identities:** $\frac{\partial x^T a}{\partial x} = a^T$, $\frac{\partial x^T B x}{\partial x} = x^T (B + B^T)$, $f = Ax \Rightarrow J = A$.
- **Chain (matrix):** $\frac{df}{dx} = \frac{\partial f}{\partial g} \frac{\partial g}{\partial x}$.
- **Sigmoid:** $\sigma' = \sigma(1 - \sigma)$; $\delta_z = \delta_a \odot a \odot (1 - a)$, $\frac{\partial L}{\partial A} = \delta_z x^T$, $\frac{\partial L}{\partial x} = A^T \delta_z$.

Links: backprop is just the chain rule on a graph.

Remember: Jacobian stacks gradients; $\sigma' = \sigma(1 - \sigma)$; state dimensions.

Taylor & Optimization L6,8

- **Taylor:** $T_n = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$; remainder $\frac{f^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$.
- **2-var:** $f(a+h, b+k) = f + (hf_x + kf_y) + \frac{1}{2}(h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}) + \dots$
- **Vector:** $f(x_0 + \delta) \approx f + \nabla f \delta + \frac{1}{2} \delta^T H \delta$; linearization = first two terms.
- **Critical:** $\nabla f = 0$; $H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix}$ symmetric.
- **Test:** PD ($f_{xx} > 0, \det H > 0$) $\rightarrow$ min; ND $\rightarrow$ max; $\det H < 0 \rightarrow$ saddle; 0 inconclusive.

Links: a positive-definite Hessian = the bridge to a guaranteed minimum.

Remember: $\nabla f = 0$ finds them; Hessian classifies: PD valley, ND hill, indefinite saddle.

One-glance sanity checks all topics

- **Eigenvalues right?** $\sum \lambda = \text{trace}$, $\prod \lambda = \det$.
- **Solution type?** compare $\text{rank} A$, $\text{rank}[A|b]$, n , $\# \text{free} = n - \text{rank}$.
- **Min vs max?** $\det H > 0$ and $f_{xx} > 0 \Rightarrow \text{min}$.
- **Shapes?** gradient $1 \times n$, Jacobian $m \times n$.
- **Valid inner product?** matrix symmetric positive-definite.

Remember: thirty seconds of checking saves whole questions.

KEEP CLOSE

Time-savers — reach for the identity, not the long way

Shortcuts that save you time

- **Triangular / diagonal:** $\det = \text{product of the diagonal}$ (don't expand).
- **Triangular:** eigenvalues are the diagonal entries.
- Read off: $\det A = \prod \lambda$, $\text{tr} A = \sum \lambda$.
- **Powers:** $\det(A^n) = (\det A)^n$, $\text{tr}(A^n) = \sum \lambda_i^n$.
- **2 x 2 eigenvalues:** $\lambda^2 - (\text{tr})\lambda + \det = 0$.
- **Orthogonal:** inverse = transpose, $Q^{-1} = Q^T$.
- **Free variables** = $n - \text{rank}$ (rank-nullity).
- A and A^T : same eigenvalues & determinant.
- **Singular:** $\det A = 0 \iff$ no inverse $\iff$ some $\lambda = 0$.
- **Singular values:** $\sigma = \sqrt{\lambda(A^T A)}$, $\|A\|_2 = \sigma_1$.
- **Modulus:** $|a + bi| = \sqrt{a^2 + b^2}$.

Special matrices at a glance

- **Symmetric** $A = A^T$: real λ , $\perp$ eigvecs.
- **Positive-definite** $x^T A x > 0$: +ve λ , Cholesky.
- **Orthogonal** $Q^T Q = I$: keeps lengths & angles.
- **Hermitian** $A^H = A$: real λ (complex case).
- **Diagonal / Triangular:** $\det = \prod \text{diagonal}$; λ on diagonal.

FOCUS YOUR REVISION

The five skills to master

- Linear systems** Row-echelon, rank, and the complete solution of $Ax = b$.
- Vector spaces** Independence, subspaces, basis & dimension.
- Eigen & geometry** Eigen-problems · inner products · Hermitian / Cholesky.
- Vector calculus** Gradient · Jacobian · chain rule (mind the dimensions).
- Optimization** Critical points · Hessian · Taylor expansion.

Master these five and you've covered the core of the course.

LAST-MINUTE RECALL

If you remember nothing else

The 10 things to never forget

1. Elimination on $Ax = b$: rank vs $\text{rank}[A|b]$ vs $n \rightarrow$ none / one / many.
2. rank = $\# \text{pivots}$ = dim of column space; pivot columns are a basis.
3. Subspace = contains **0** and closed under + and scaling.
4. $\langle x, y \rangle = x^T A y$ needs SPD; $\cos \theta = \text{dot} \div \text{norms}$.
5. Eigen: $\det(A - \lambda I) = 0$; eigenspace = $\text{null}(A - \lambda I)$.
6. trace = $\sum \lambda$, $\det = \prod \lambda$; triangular $\Rightarrow \lambda$ on diagonal.
7. SVD: V from $A^T A$, U from AA^T , $\sigma = \sqrt{\text{eig}}$, $\|A\|_2 = \sigma_1$.
8. Gradient = row $1 \times n$; Jacobian $m \times n$; $f = Ax \Rightarrow J = A$.
9. Backprop = chain rule run backward; $\sigma' = \sigma(1 - \sigma)$.
10. $\nabla f = 0$, then Hessian: PD $\rightarrow$ min, ND $\rightarrow$ max, $\det H < 0 \rightarrow$ saddle.